

PostgreSQL pg_hba.conf

PostgreSQL pg_hba.conf Guide

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1. What is pg_hba.conf?

HBA = Host-Based Authentication.

Location: usually inside PostgreSQL data directory

(/var/lib/pgsql/<version>/data/pg_hba.conf or
/pgData/pgsql15/data/pg_hba.conf depending on your setup).

Format:

TYPE DATABASE USER ADDRESS METHOD

TYPE: local, host, hostssl, hostnossl

DATABASE: which DBs (e.g. all, postgres, mydb)

USER: which roles (e.g. all, myuser)

ADDRESS: client IP/CIDR (127.0.0.1/32, 192.168.2.0/24)

METHOD: authentication method (trust, md5, scram-sha-256, peer, reject, cert)

OPTIONS : optional settings (e.g., clientcert=1)

2. Authentication Methods in pg_hba.conf

trust:

- > No password needed. Anyone who can connect to the server is trusted.
- > Use: testing only.
- > Not secure in production.

Example:

```
host      all      all      127.0.0.1/32      trust
```

md5

- > Password required, stored as MD5 hash.
- > Legacy support. Weaker security, avoid if possible.

Example:

```
host      all      all      127.0.0.1/32      md5
```

scram-sha-256

- > Password required, stored as salted SCRAM-SHA-256 hash.
- > Stronger than MD5.
- > Recommended method for production.

Example:

```
host      all      all      127.0.0.1/32      scram-sha-256
```

peer

- > Works for local connections only.
- > The OS user name must match the PostgreSQL role.
- > Good for local scripts/services under same user.

Example:

```
local    all    all                                peer
```

reject

- > Explicitly denies access.
- > Useful for blocking unwanted connections (like a firewall rule).

Example:

```
host     all    all    0.0.0.0/0        reject
```

cert

- > Requires SSL/TLS client certificate.
- > PostgreSQL role must match certificate username.
- > Very secure for enterprise / production with SSL.

Example:

```
hostssl all    all    192.168.1.0/24    cert clientcert=1
```

3. Create Database and User

```
postgres=# CREATE USER teja WITH PASSWORD 'teja123';
CREATE ROLE
postgres=# CREATE DATABASE orcl OWNER teja;
```

```
CREATE DATABASE
postgres=#
```

4. Contents of pg_hba.conf

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf
# TYPE      DATABASE          USER            ADDRESS
METHOD

# Allow local peer access for postgres
local       all              postgres
peer

# Allow local password access for all
local       all              all
scram-sha-256

[postgres@pg17 ~]$
```

5. Play with peer

Works for local connections only. The OS user name must match the PostgreSQL role.

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf
# TYPE      DATABASE          USER            ADDRESS
METHOD

# Allow local peer access for postgres

local       all              postgres
peer

# Allow local password access for all
local       all              all
scram-sha-256
```

```
[postgres@pg17 ~]$  
  
[postgres@pg17 ~]$ psql -U teja -d orcl  
Password for user teja: <---- it's asking password  
  
psql (15.13)  
Type "help" for help.  
  
orcl=> exit  
[postgres@pg17 ~]$  
  
Password Not asking for user postgres  
[postgres@pg17 ~]$ psql -U postgres -d orcl  
psql (15.13)  
Type "help" for help.  
  
orcl=#
```

6. Play with trust

No password needed. Anyone who can connect to the server is trusted.

Use: testing only. Not secure in production.

Change pg_hba.conf:

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf  
# TYPE      DATABASE          USER            ADDRESS  
METHOD  
  
# Allow local peer access for postgres  
local      all                postgres  
peer  
  
# Allow local password access for all  
  
local      all                all  
trust  
  
[postgres@pg17 ~]$
```

```
[postgres@pg17 ~]$ /usr/pgsql-15/bin/pg_ctl reload -D
/pgData/pgsql15/data/
server signaled
[postgres@pg17 ~]$ psql -U teja -d orcl <--- It won't ask for
a password.
psql (15.13)
Type "help" for help.

orcl=>
```

7. Play with reject

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf
# TYPE      DATABASE          USER            ADDRESS
METHOD

# Allow local peer access for postgres
local      all                postgres
peer

# Allow local password access for all

local      all                all
reject

[postgres@pg17 ~]$

[postgres@pg17 ~]$ /usr/pgsql-15/bin/pg_ctl reload -D
/pgData/pgsql15/data/
server signaled
[postgres@pg17 ~]$ psql -U teja -d orcl

psql: error: connection to server on socket
"/run/postgresql/.s.PGSQL.5432" failed: FATAL:  pg_hba.conf
rejects connection for host "[local]", user "teja", database
"orcl", no encryption

[postgres@pg17 ~]$
```

8. Restrict by User

Allow only user teja, block everyone else:

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf
# TYPE      DATABASE          USER            ADDRESS
METHOD

# Allow local peer access for postgres
local      all                postgres
peer

# Allow local password access for all
local      all                all
trust

# Allow IPv4 localhost

host       orcl              teja            192.168.2.31/32
scram-sha-256
host       orcl              all             192.168.2.31/32
reject

[postgres@pg17 ~]$

[postgres@pg17 ~]$ /usr/pgsql-15/bin/pg_ctl reload -D
/pgData/pgsql15/data/
server signaled
[postgres@pg17 ~]$ psql -U teja -d orcl -h 192.168.2.31
Password for user teja:
psql (15.13)
Type "help" for help.

orcl=> \conninfo
You are connected to database "orcl" as user "teja" on host
"192.168.2.31" at port "5432".
orcl=>
orcl=> exit
[postgres@pg17 ~]$
```

```
[postgres@pg17 ~]$ psql -U postgres -d orcl -h 192.168.2.31
```

```
psql: error: connection to server at "192.168.2.31", port 5432
failed: FATAL:  pg_hba.conf rejects connection for host
"192.168.2.31", user "postgres", database "orcl", no
encryption
```

```
[postgres@pg17 ~]$
```

Notice: Only user Teja able to connect where as user postgres not able to connect

9. Restrict by Network

We want to block connections from IP 192.168.2.0/24

```
[postgres@pg17 ~]$ cat /pgData/pgsql15/data/pg_hba.conf
```

```
# TYPE  DATABASE          USER              ADDRESS
METHOD
```

```
# Allow local peer access for postgres
local   all                postgres
peer
```

```
# Allow local password access for all
local   all                all
trust
```

```
# Allow IPv4 localhost
```

```
host    all                all                192.168.2.0/24
reject
```

```
[postgres@pg17 ~]$
```

```
[postgres@pg17 ~]$ /usr/pgsql-15/bin/pg_ctl reload -D
/pgData/pgsql15/data/
```



```

server signaled
[postgres@pg17 ~]$
[postgres@pg17 ~]$ psql -U teja -d orcl -h 192.168.2.31

psql: error: connection to server at "192.168.2.31", port 5432
failed: FATAL:  pg_hba.conf rejects connection for host
"192.168.2.31", user "teja", database "orcl", no encryption

[postgres@pg17 ~]$

```

10. Verify logs

```

postgres=# SELECT * FROM pg_hba_file_rules;
 line_number | type   | database | user_name | address      |
netmask      | auth_method | options | error
-----+-----+-----+-----+-----+-----
-----+-----+-----+-----+-----+-----
          4 | local  | {all}    | {postgres} |              |
| peer      |          |          |            |              |
          7 | local  | {all}    | {all}      |              |
| trust     |          |          |            |              |
         11 | host   | {orcl}   | {teja}     | 192.168.2.31 |
255.255.255.255 | scram-sha-256 |          |            |              |
         12 | host   | {orcl}   | {all}      | 192.168.2.31 |
255.255.255.255 | reject          |          |            |              |
(4 rows)

postgres=#

[postgres@pg17 ~]$ tail -f
/pgData/pgsql15/data/log/postgresql-Tue.log
2025-09-16 04:08:37.244 EDT [8294] LOG:  database system was
shut down at 2025-09-16 04:08:32 EDT
2025-09-16 04:08:37.251 EDT [8289] LOG:  database system is
ready to accept connections
2025-09-16 04:11:27.459 EDT [8289] LOG:  received SIGHUP,
reloading configuration files
2025-09-16 04:11:31.842 EDT [8401] FATAL:  pg_hba.conf rejects
connection for host "[local]", user "teja", database "orcl",
no encryption

```

```
2025-09-16 04:13:37.568 EDT [8292] LOG:  checkpoint starting:
time
2025-09-16 04:13:37.573 EDT [8292] LOG:  checkpoint complete:
wrote 3 buffers (0.0%); 0 WAL file(s) added, 0 removed, 0
recycled; write=0.002 s, sync=0.002 s, total=0.006 s; sync
files=2, longest=0.001 s, average=0.001 s; distance=0 kB,
estimate=0 kB
2025-09-16 04:16:15.863 EDT [8289] LOG:  received SIGHUP,
reloading configuration files
2025-09-16 04:16:52.318 EDT [8474] FATAL:  pg_hba.conf rejects
connection for host "192.168.2.31", user "teja", database
"orcl", no encryption
2025-09-16 04:27:15.932 EDT [8289] LOG:  received SIGHUP,
reloading configuration files
2025-09-16 04:27:50.593 EDT [8613] FATAL:  pg_hba.conf rejects
connection for host "192.168.2.31", user "postgres", database
"orcl", no encryption
```

Caution: *Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.*

Thank you,

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